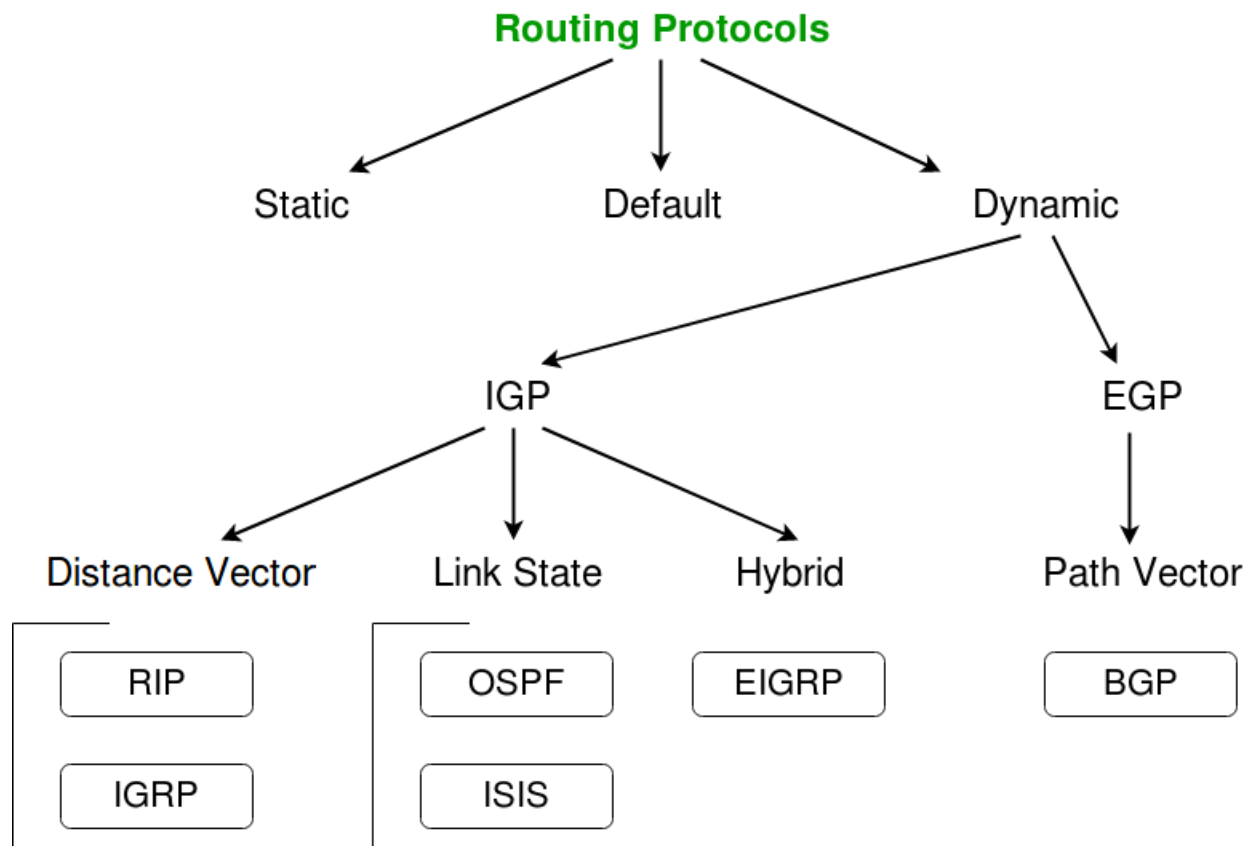
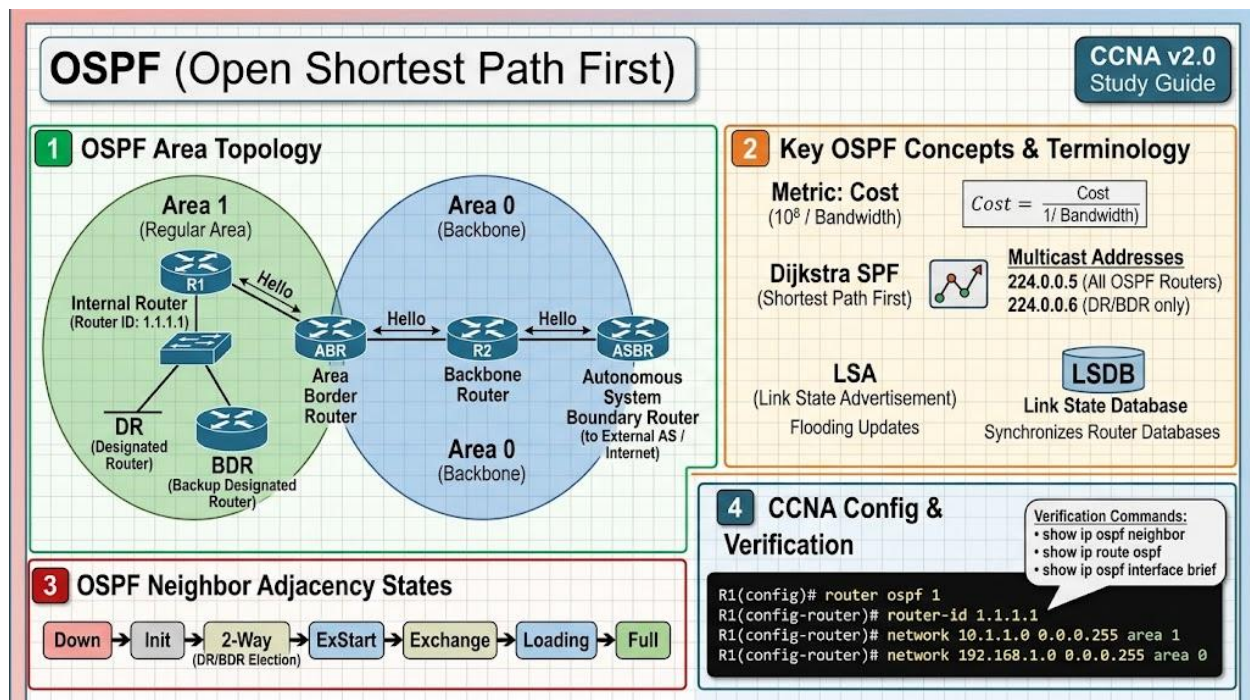
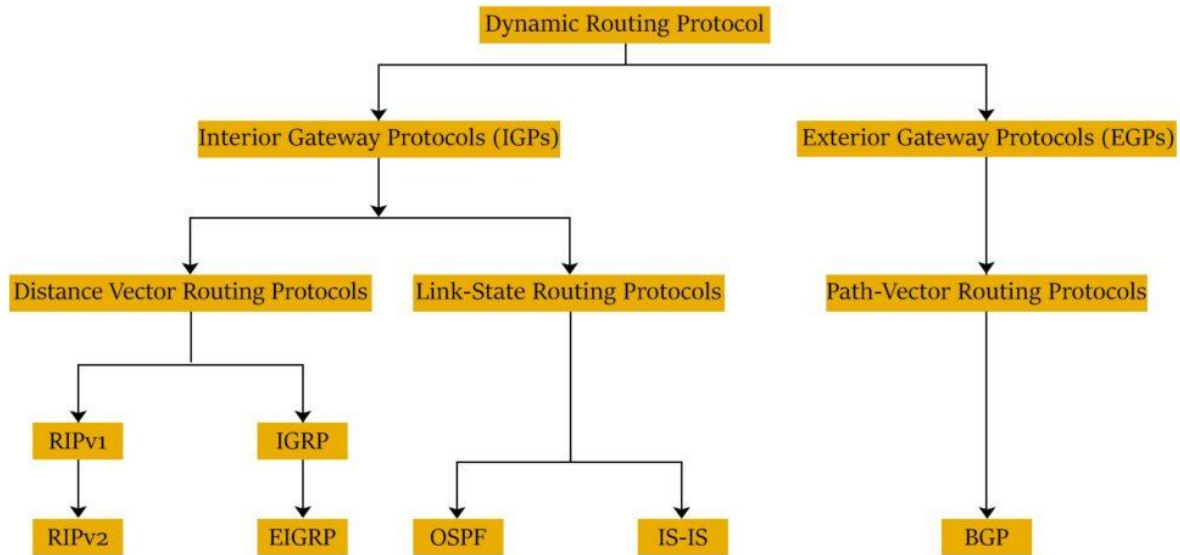


OSPF (Open Shortest Path First)





OSPF is a standardized Link-State interior gateway routing protocol (IGP) widely used in enterprise networks. It uses Dijkstra's Shortest Path First (SPF) algorithm to calculate the most efficient, loop-free path to each destination network.

Metric: OSPF uses **Cost** as its metric, which is inversely proportional to interface bandwidth (Reference formula: $\text{Cost} = 100,000,000 / \text{Bandwidth in bps}$). Lower cost indicates a preferred path.

Administrative Distance (AD): 110.

Communication: Uses Multicast IPs **224.0.0.5** (All OSPF Routers) and **224.0.0.6** (DR/BDR only) to share routing updates dynamically.

OSPF Data Structures

To maintain a complete map of the network, OSPF builds and maintains three distinct tables:

- **Neighbor Table (Adjacency Database):** Contains a list of all recognized neighboring OSPF routers that share a direct link.
- **Topology Table (Link State Database - LSDB):** A complete map of the network topology built from Link State Advertisements (LSAs) flooded by all routers in the area. All routers in the same area have identical LSDBs.
- **Routing Table (Forwarding Database):** Contains the best, lowest-cost paths to every known destination, calculated by running the SPF algorithm against the LSDB.

Network Roles and Areas

OSPF uses a hierarchical design based on "Areas" to minimize routing update traffic and reduce memory and CPU overhead on individual routers.

- **Router ID (RID):** The unique identifier for an OSPF router. It defaults to the highest active Loopback IP address, followed by the highest active physical interface IP address.
- **DR (Designated Router):** Elected on multi-access networks (like Ethernet) to act as a central point of contact for LSA distribution, minimizing excessive update traffic.
- **BDR (Backup Designated Router):** Passively monitors the DR and immediately takes over if the DR fails.
- **Area 0 (Backbone Area):** The core of the OSPF network. All other regular areas must connect directly to Area 0.
- **ABR (Area Border Router):** A router with interfaces in multiple areas, acting as the gateway between regular areas and Area 0.
- **ASBR (Autonomous System Boundary Router):** A router that connects the OSPF routing domain to an external network or a different routing protocol (e.g., BGP, EIGRP).

Neighbor Adjacency States

OSPF routers must successfully transition through seven distinct states to establish a fully synchronized neighbor adjacency. This process ensures both routers agree on network parameters and have identical maps of the network before routing traffic.

- **Down State:** The initial state. No OSPF Hello packets have been received from the neighbor. The router is active and sending its own Hello packets out of its OSPF-enabled interfaces, waiting for a response.
- **Init State:** A Hello packet has been received from another router, but bi-directional communication is not yet established. The receiving router sees the neighbor's Hello, but its own Router ID (RID) is not yet listed in the neighbor's Hello packet.
- **2-Way State:** Bi-directional communication is successfully established. The router receives a Hello packet containing its own RID in the neighbor list.
 - *Crucial Event:* On multi-access networks (like Ethernet), the Designated Router (DR) and Backup Designated Router (BDR) election takes place at the end of this state.
 - *Note:* Routers that are neither DR nor BDR (called DROthers) will remain in the 2-Way state with each other permanently to save bandwidth. They only form full adjacencies with the DR and BDR.
- **ExStart (Exchange Start) State:** The routers prepare to synchronize their Link State Databases (LSDB). They negotiate a Master/Slave relationship to determine who dictates the flow of information. The router with the highest RID becomes the Master.
- **Exchange State:** The routers trade Database Description (DBD) packets. These packets do not contain full routing information; instead, they act as a table of contents, providing a high-level summary of the LSA headers contained in each router's LSDB.
- **Loading State:** If a router notices a newer or missing route during the Exchange state, it enters the Loading state. It sends Link State Requests (LSRs) asking for the complete details of specific networks. The neighbor responds with Link State Updates (LSUs) containing the full Link State Advertisements (LSAs).
- **Full State:** The routers have completely synchronized their databases. They share an identical LSDB and are fully adjacent. Normal OSPF routing calculations (running the SPF algorithm) now occur, and the routers will only exchange incremental updates if the network topology changes.

OSPF Metric

OSPF calculates its metric, known as **Cost**.

The Cost Formula

The fundamental formula used by OSPF to compute the cost of an individual interface is:

$$\text{Cost} = \frac{\text{Reference Bandwidth}}{\text{Interface Bandwidth}}$$

- **Reference Bandwidth:** A configurable baseline value defined within OSPF, defaulting to **10⁸ bps (100 Mbps)**.
- **Interface Bandwidth:** The actual operating speed of the interface (e.g., 10 Mbps, 100 Mbps, 1 Gbps, 10 Gbps).

Because the default reference bandwidth is 100 Mbps, any link running at 100 Mbps or faster will result in a calculated cost of less than 1, which rounds down to **1** in integer arithmetic. This creates a scalability bottleneck in modern networks, as Fast Ethernet (100 Mbps), Gigabit Ethernet (1 Gbps), and 10-Gigabit Ethernet interfaces would all share the exact same minimum cost of 1.

Step-by-Step Calculation Examples

1. Using Default Reference Bandwidth (100 Mbps / 10⁸)

- **Fast Ethernet (100 Mbps / 10⁸ bps):**

$$\text{Cost} = \frac{100,000,000}{100,000,000} = 1$$

- **Gigabit Ethernet (1 Gbps / 10⁹ bps):**

$$\text{Cost} = \frac{100,000,000}{1,000,000,000} = 0.1 \rightarrow \text{rounds to } 1$$

- **10-Gigabit Ethernet (10 Gbps / 10¹⁰ bps):**

$$\text{Cost} = \frac{100,000,000}{10,000,000,000} = 0.01 \rightarrow \text{rounds to } 1$$

2. Cumulative Path Cost

The total metric to reach a destination network is the **sum of the exit interface costs** along the path from the source router to the destination router.

- If a packet exits Router A through a Fast Ethernet interface (Cost = 1) and crosses a link to enter Router B through a Gigabit interface (Cost = 1), the total cumulative cost for that leg of the path is $1 + 1 = 2$.

Modifying the Reference Bandwidth

To differentiate between modern high-speed links (like 1 Gbps and 10 Gbps), network engineers typically increase the reference bandwidth so that faster links receive granular, distinct cost values.

The command to change it (e.g., setting it to 100 Gbps / 10^{11}) is executed in OSPF configuration mode:

Cisco CLI

```
Router(config)# router ospf 1
```

```
Router(config-router)# auto-cost reference-bandwidth 100000
```

(Note: Reference bandwidth is specified in Mbps, so 100,000 Mbps equals 100 Gbps).

When a reference bandwidth of 100 Gbps is applied, the calculations change:

- **Fast Ethernet (100 Mbps):** $\frac{100,000,000}{100,000} = 1000$
- **Gigabit Ethernet (1 Gbps):** $\frac{100,000,000}{1,000,000} = 100$
- **10-Gigabit Ethernet (10 Gbps):** $\frac{100,000,000}{10,000,000} = 10$

Important Rule: If you change the reference bandwidth, you **must change it on all OSPF routers** in the routing domain. If mismatched, routers will calculate inconsistent path metrics, leading to suboptimal routing loops or black-holed traffic. Alternatively, you can manually override an interface's cost using the **ip ospf cost [value]** command directly on the interface configuration level without modifying the global reference bandwidth.

```
Router(config)# interface GigabitEthernet0/0
```

```
Router(config-if)# ip ospf cost 50
```

Basic Configuration Example:

```
Router(config)# router ospf 1
```

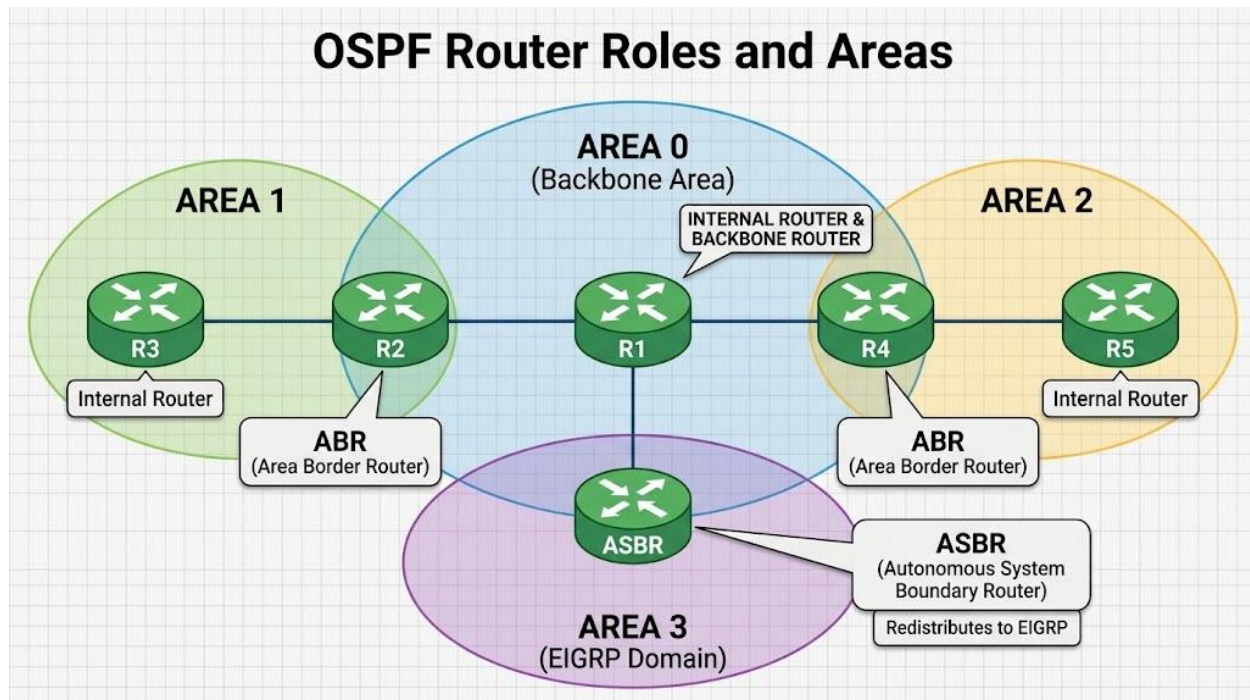
```
Router(config-router)# router-id 1.1.1.1
```

```
Router(config-router)# network 10.1.1.0 0.0.0.255 area 0
```

```
Router(config-router)# network 192.168.10.0 0.0.0.255 area 1
```

Link State Advertisement (LSA)(MOST Important)

- **OSPF Basics:** A link-state routing protocol that utilizes the SPF (Shortest Path First) algorithm to calculate the optimal path between a source and destination.
- **Database Synchronization:** OSPF routers exchange Link State Advertisements (LSAs) to construct and maintain an accurate, shared map of the network topology.
- **Prerequisite for LSAs:** Before analyzing the different types of LSAs, it is essential to first understand the specific roles routers play within an OSPF network.



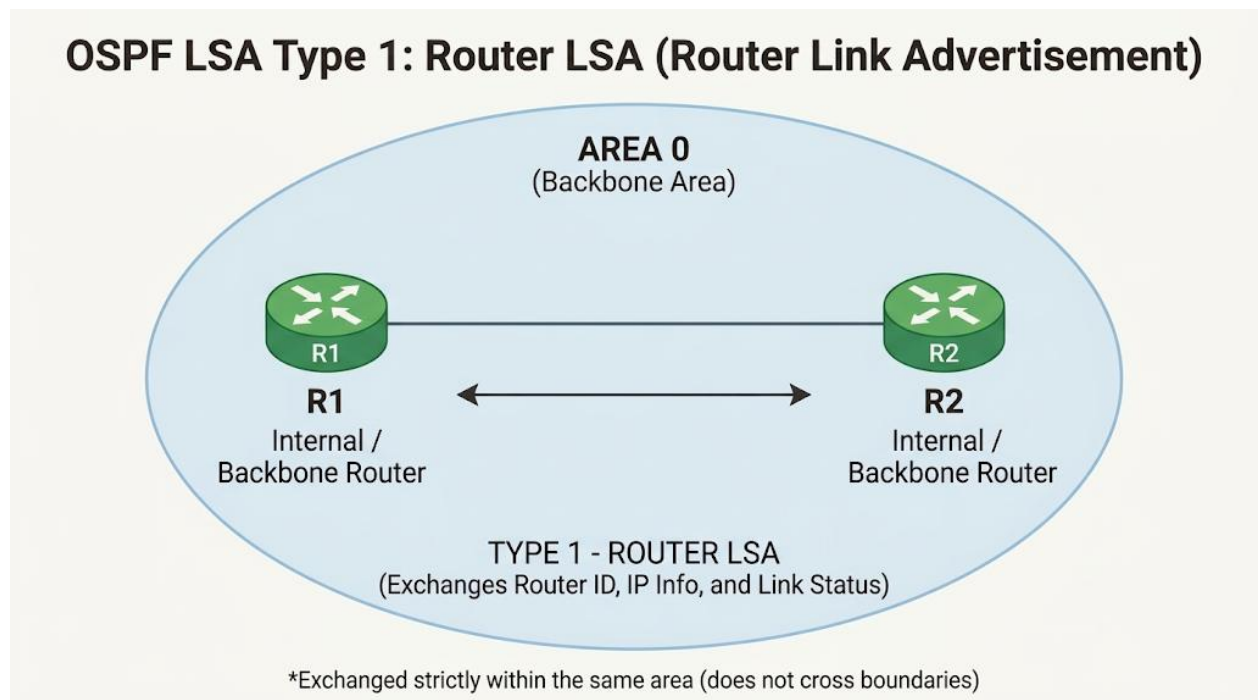
1. **Backbone router** - The area 0 is known as backbone area and the routers in area 0 are known as backbone routers.
2. **Internal router** - An internal router is a router which have all of its interfaces in a single area.
3. **Area Boundary Router (ABR)** - The router which connects backbone area with another area is called Area Boundary Router. The ABRs therefore maintain multiple link-state databases that describe both the backbone topology and the topology of the other areas.
4. **Autonomous System Border Router (ASBR)** - When an OSPF router is connected to a different protocol like EIGRP, or Border Gateway Protocol, or any other routing protocol then it is known as AS. The router which connects two different AS (in which one of the interface is operating OSPF in area 0) is known as Autonomous System Border Router. These routers perform redistribution. ASBRs run both OSPF and another routing protocol, such as RIP or BGP.

LSA Types-

There are different types of LSAs exchanged depending upon the areas in which devices operating OSPF reside.

LSA Type 1: Router Link Advertisement (Router LSA)

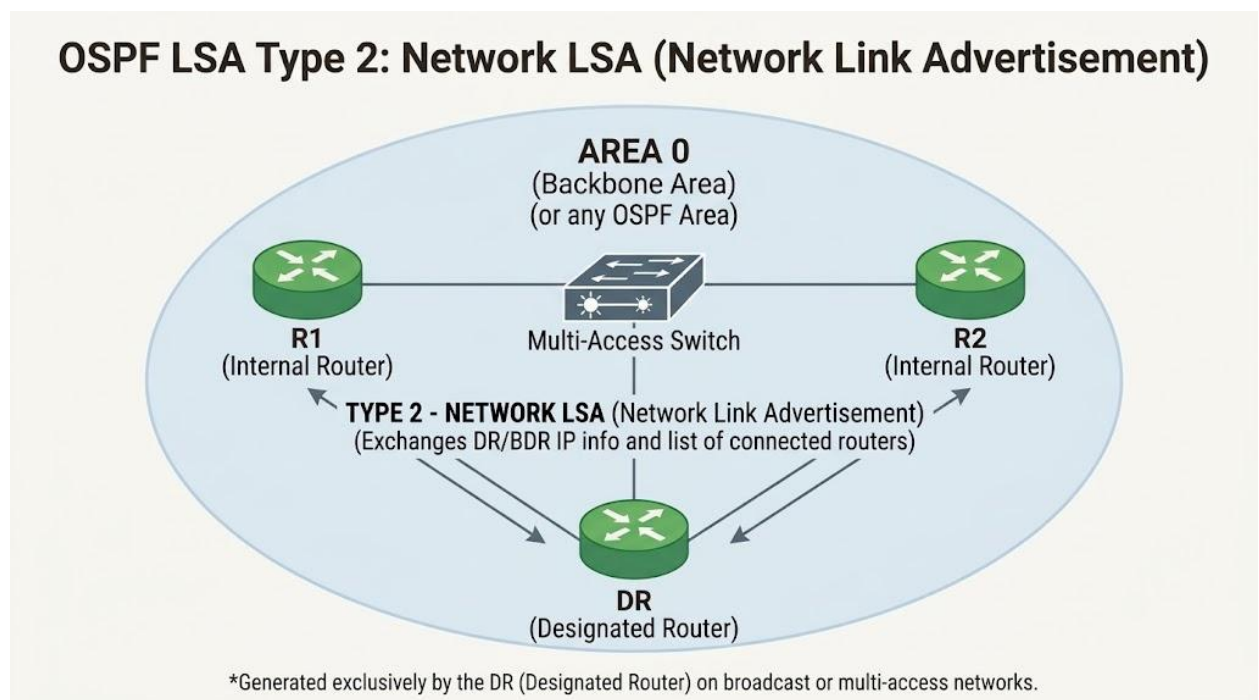
- **Scope:** Exchanged exclusively among routers within the same OSPF area.
- **Contents:** Advertises the originating router's specific details, including its Router ID, active link statuses, IP information, and current interface states.
- **Multi-Area Behavior:** If a router spans multiple areas (e.g., an Area Border Router), it must generate and flood a separate, distinct Type 1 LSA into each individual area it connects to.



As shown in the figure, Type 1 LSA are exchanged by the routers within the same area but if other interface of router is in another area then different Type 1 LSA will be exchanged.

LSA Type 2: Network Link Advertisement (Network LSA)

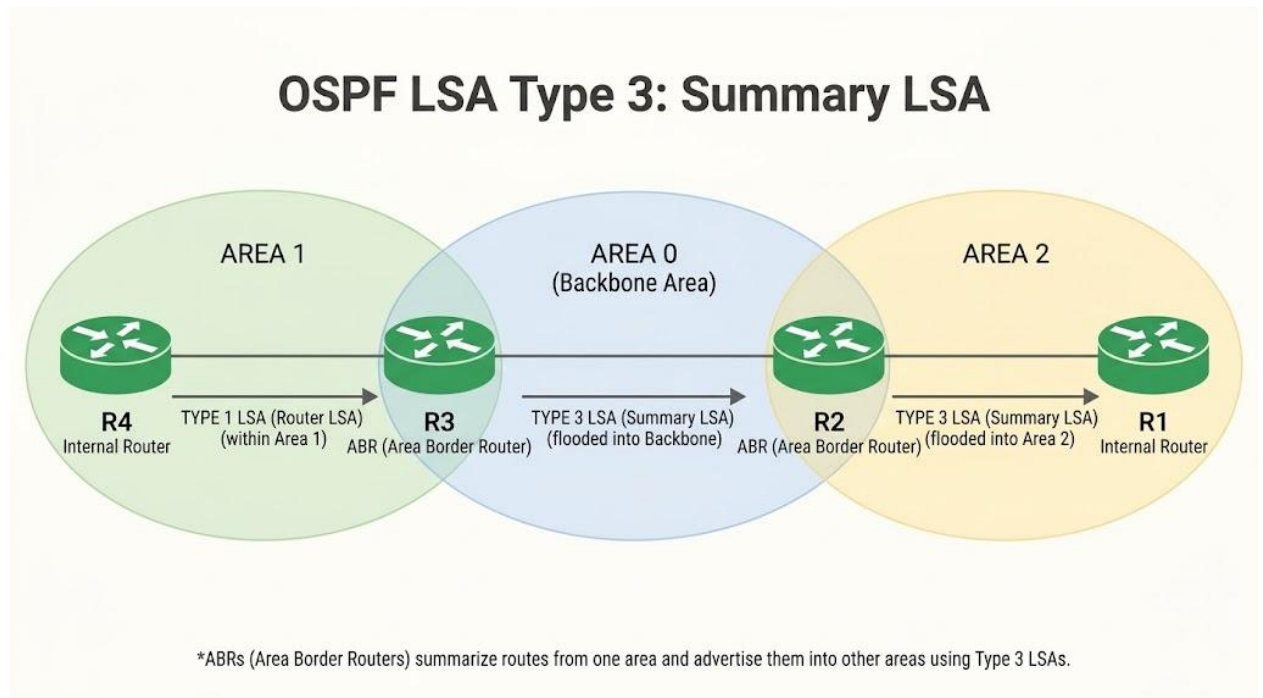
- **Generated By:** Exclusively created and sent by the Designated Router (DR) on broadcast or multi-access networks (e.g., Ethernet).
- **Contents:** Advertises the IP information of both the DR and the Backup Designated Router (BDR), alongside the states of all other routers connected to that specific network segment.
- **Function:** Centralizes routing updates. In a broadcast network, the DR acts as the single source responsible for distributing routing information to all other routers on that segment.
- **Scope:** Flooded strictly to routers within the same local OSPF area.



LSA Type 3: Summary LSA

- **Generated By:** Area Border Routers (ABRs).

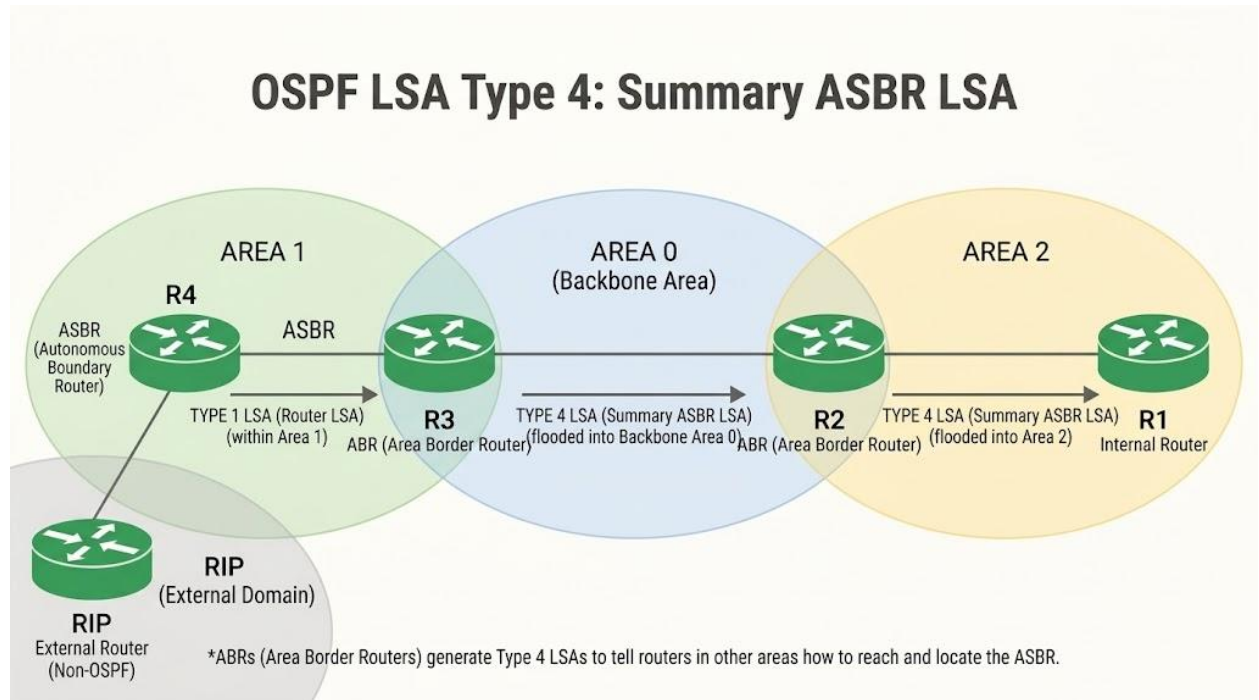
- **Function:** Acts as a messenger between different OSPF areas. The ABR takes the topological database it learns from one area (like Area 1) and injects that routing information into another area (such as the backbone Area 0).
- **Contents:** Contains the advertised IP network information as well as the Router ID of the ABR that is generating the LSA.
- **Scope:** Crosses area boundaries to facilitate inter-area routing (flooding routes from one area to another).



LSA Type 4: Summary ASBR LSA

- **Generated By:** Area Border Routers (ABRs).
- **Function:** Provides routers in other areas with the routing path (directions) required to reach an Autonomous System Boundary Router (ASBR).
- **Mechanism:** An ASBR first advertises its presence within its local area using a standard Type 1 LSA. When the ABR receives this Type 1 LSA, it generates a Type 4 LSA and floods

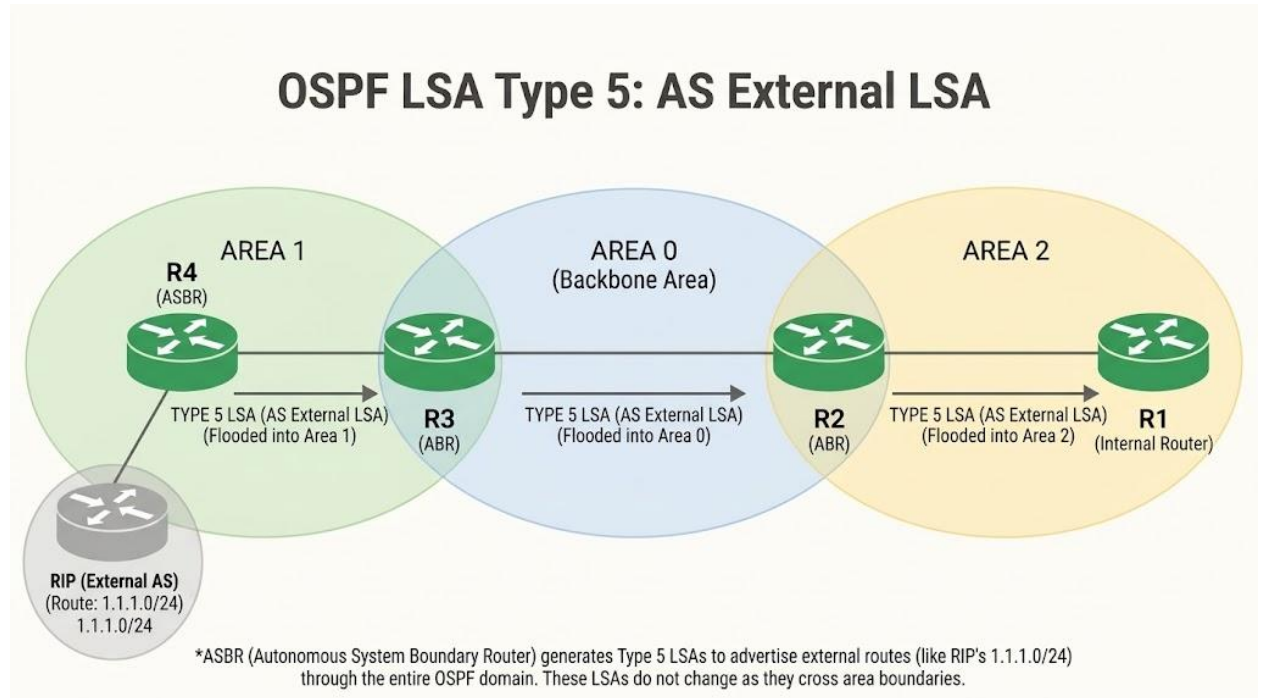
it into other areas, ensuring all routers across the OSPF domain know how to route traffic to that specific ASBR.



LSA Type 5: AS External Link Advertisement

- **Generated By:** Autonomous System Boundary Routers (ASBRs).
- **Function:** Advertises external routes originating from outside the OSPF routing domain, such as networks learned from other routing protocols (like RIP or EIGRP) or static routes to the Internet.
- **Mechanism & Scope:** When an ASBR connects to a foreign network, it is responsible for translating those external routes into Type 5 LSAs. These LSAs are then flooded

throughout all normal areas of the entire OSPF network to ensure every router knows how to reach the external destinations.



LSA Type 7: NSSA External LSA

- **Generated By:** Autonomous System Boundary Routers (ASBRs) located specifically within a **Not-So-Stubby Area (NSSA)**.
- **Function:** Advertises external routes (like those from RIP, EIGRP, or the Internet) into an NSSA. Because strict "Stub" areas do not allow standard Type 5 External LSAs, the Type 7 LSA acts as a special workaround to bring external routes into this specific area type.
- **Mechanism & Scope:** The ASBR generates the Type 7 LSA, which is flooded *only* within the local NSSA. When this Type 7 LSA reaches the Area Border Router (ABR), the ABR translates it into a standard Type 5 LSA and floods it into the Backbone (Area 0) and the rest of the OSPF network.

OSPF LSA Type 7: NSSA External LSA

